
```
helm version
```

```
helm repo add csi-driver-nfs https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/master/charts
```

```
helm repo list
```

```
helm repo update
```

```
helm install csi-driver-nfs csi-driver-nfs/csi-driver-nfs --namespace kube-system
```

Verify the NFS CSI Driver pods:

```
kubectl get pods -n kube-system | grep nfs
```

Watch the CSI Driver pods:

```
kubectl --namespace=kube-system get pods --  
selector="app.kubernetes.io/instance=csi-driver-nfs" --watch
```

Alternative to the Helm chart:

```
curl -skSL https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/v4.13.2/deploy/install-driver.sh | bash -s v4.13.2 --
```

Note: The above command is optional and can be used instead of the Helm chart installation method.

NFS Server

```
nfs-server.yaml
```

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nfs-server
```

```
  namespace: poc
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nfs-server
  template:
    metadata:
      labels:
        app: nfs-server
    spec:
      containers:
        - name: nfs-server
          image: itsthenetwork/nfs-server-alpine:latest
          securityContext:
            privileged: true
          env:
            - name: SHARED_DIRECTORY
              value: /exports
          ports:
            - name: nfs
              containerPort: 2049
              protocol: TCP
          volumeMounts:
            - name: exports
              mountPath: /exports
      volumes:
        - name: exports
          emptyDir: {}
```

```
apiVersion: v1
kind: Service
metadata:
  name: nfs-server
  namespace: poc
spec:
  type: ClusterIP
  selector:
    app: nfs-server
  ports:
    - name: nfs
      port: 2049
      targetPort: 2049
      protocol: TCP
```

NFS StorageClass

```
avej@DESKTOP-VK19M09:POC-8$ cat nfs-storageclass.yaml
```

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: nfs-csi
provisioner: nfs.csi.k8s.io
parameters:
  server: nfs-server.poc.svc.cluster.local
  share: /
  onDelete: delete
mountOptions:
  - nfsvers=4.1
```

Shared PersistentVolumeClaim

```
avej@DESKTOP-VK19M09:POC-8$ cat shared-pvc.yaml
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: shared-data
  namespace: poc
spec:
  accessModes:
    - ReadWriteMany
  storageClassName: nfs-csi
  resources:
    requests:
      storage: 1Gi
```

RWX Test Deployment

```
avej@DESKTOP-VK19M09:POC-8$ cat rwx-test.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: rwx-test
  namespace: poc
spec:
  replicas: 2
  selector:
    matchLabels:
      app: rwx-test
  template:
    metadata:
      labels:
        app: rwx-test
    spec:
      containers:
        - name: test
          image: busybox:1.36
          command:
            - sh
            - -c
            - |
              echo "Pod started: $(hostname)"
              sleep 3600
          volumeMounts:
            - name: shared
              mountPath: /shared
      volumes:
        - name: shared
          persistentVolumeClaim:
            claimName: shared-data
```

```
avej@DESKTOP-VK19M09:POC-8$
```

Verify Shared Storage Between Two Pods

Write data from the first pod:

```
kubectl exec -n poc rwx-test-67b545d787-8ngkz -- sh -c 'echo "hello from $(hostname)" > /shared/test.txt'
```

Read the same data from the second pod:

```
kubectl exec -n poc rwx-test-67b545d787-v8n2w -- cat /shared/test.txt
```
